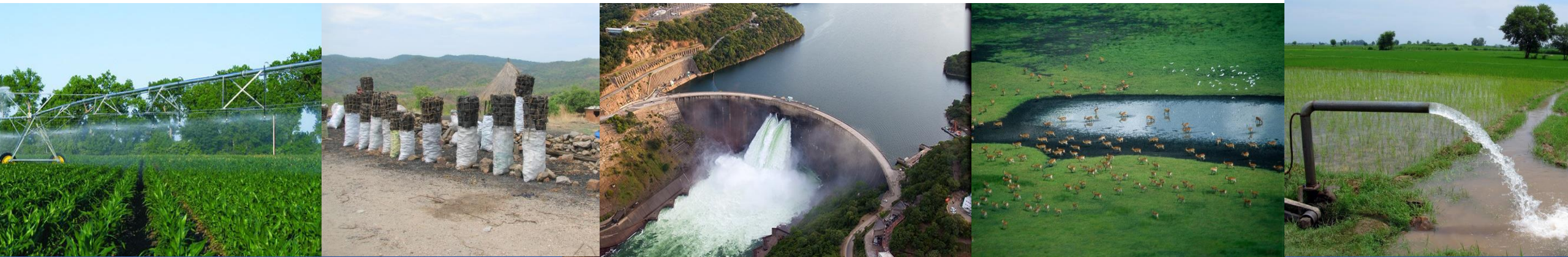


Lakes and Reservoirs

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Water Program



Reservoirs and lakes

1. Standard CWatM / methodology
2. CWatM New: updates on Reservoir module
3. Exercise on Standard CWatM
4. Exercise setting up files for running new version of CWatM

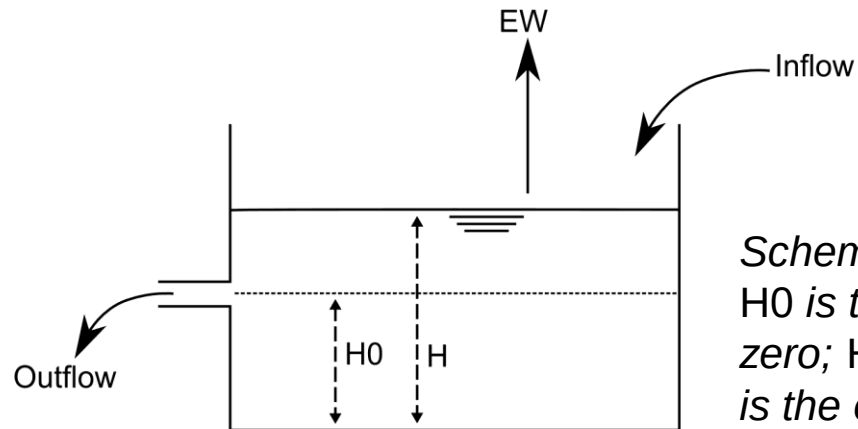


Lakes

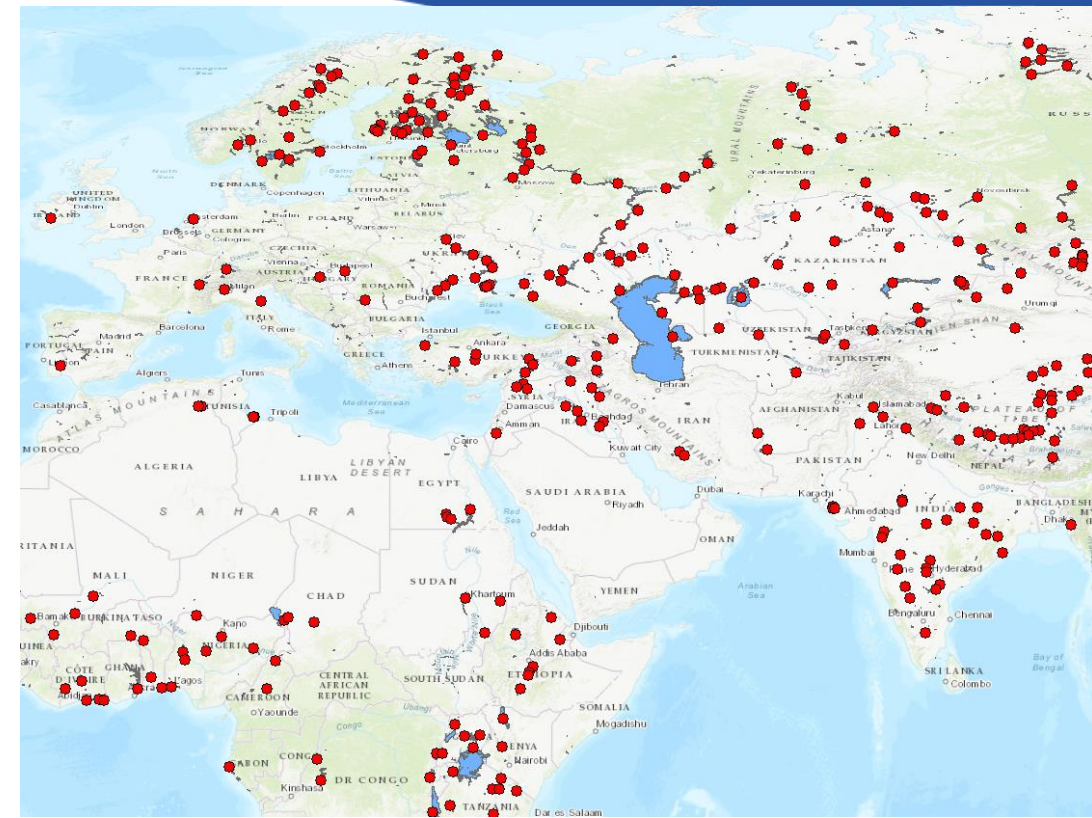
Outflow is computed using this equation:

$$O_{lake} = A(H - H_0)^B$$

O_{lake}	: Lake outflow rate [$\text{m}^3 \text{s}^{-1}$]
H	: Water level in lake [m]
H_0	: Water level at which lake outflow is zero [m]
A, B	: Constants [-]

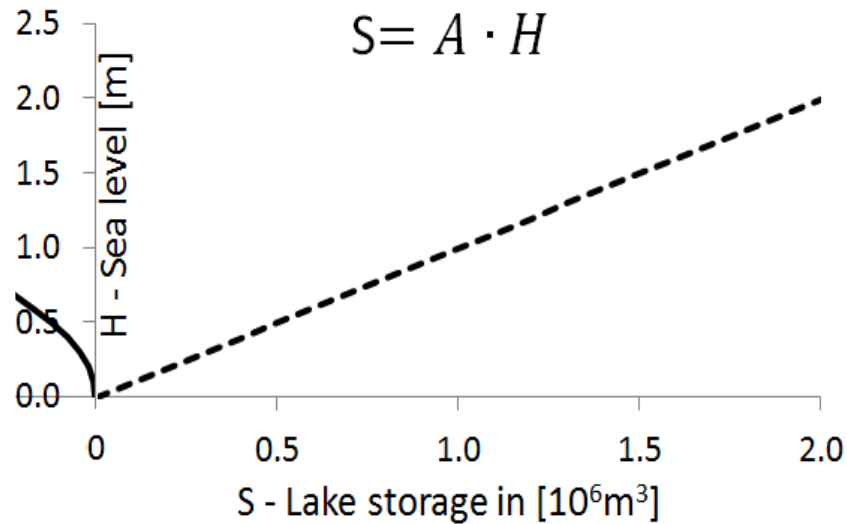


Schematic overview of the simulation of lakes. H_0 is the water level at which the outflow is zero; H is the water level in the lake and EW is the evaporation from the lake

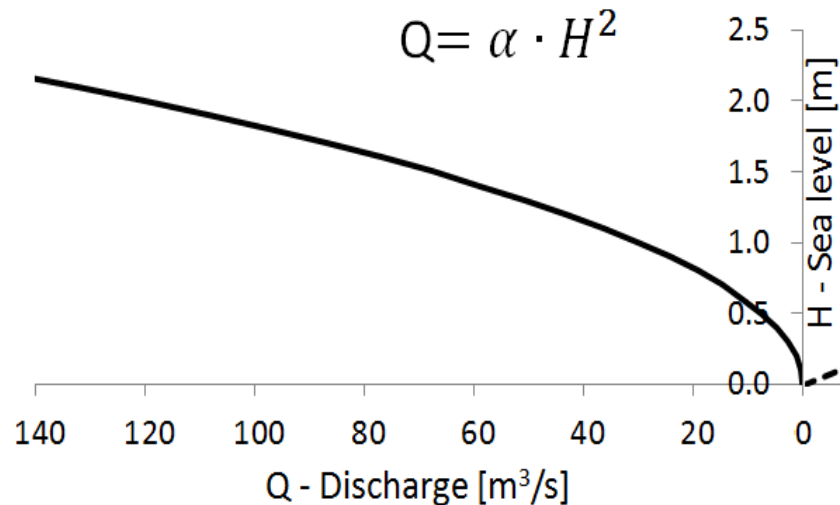


Lakes and reservoirs in CWatM

Lakes



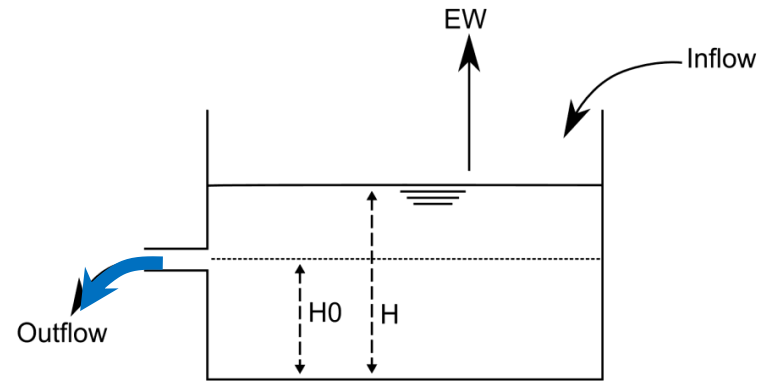
Relation between sea level and lake storage
 - linear (assuming the increase in area is neglectable)



Relation between sea level and lake outflow

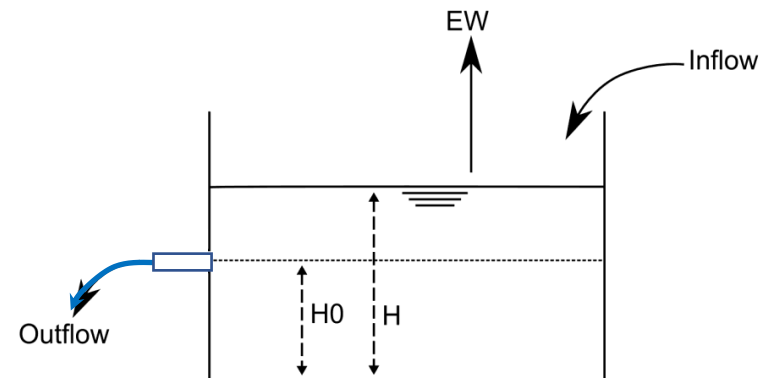
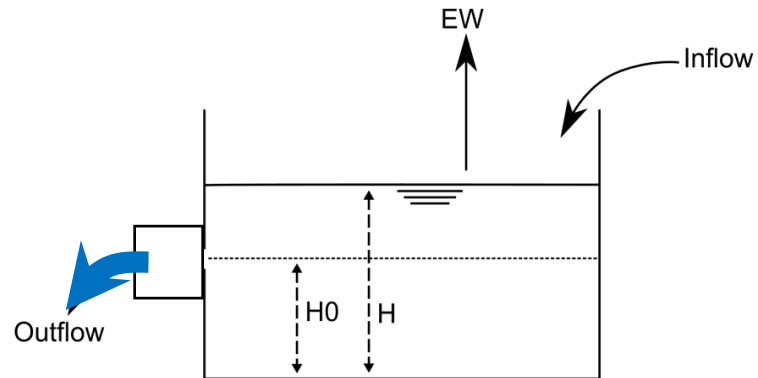
- non linear (weir formular)
- α is a function of outlet width
- outlet width is a function of discharge at outlet

Lakes – outlet width

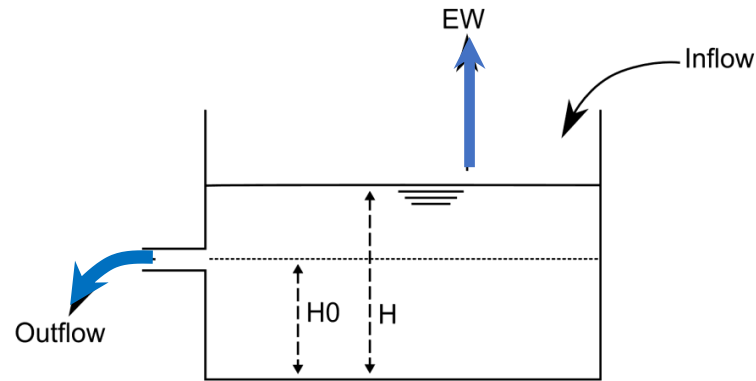


Increasing outlet width
- Increasing outflow

Decreasing outlet width
- Decreasing outflow



Lakes – lake area

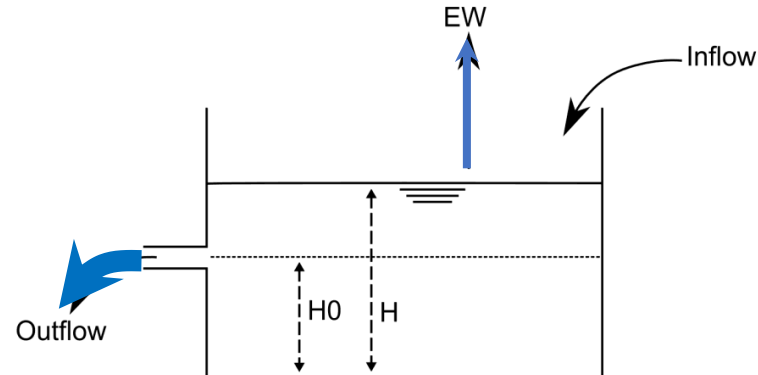
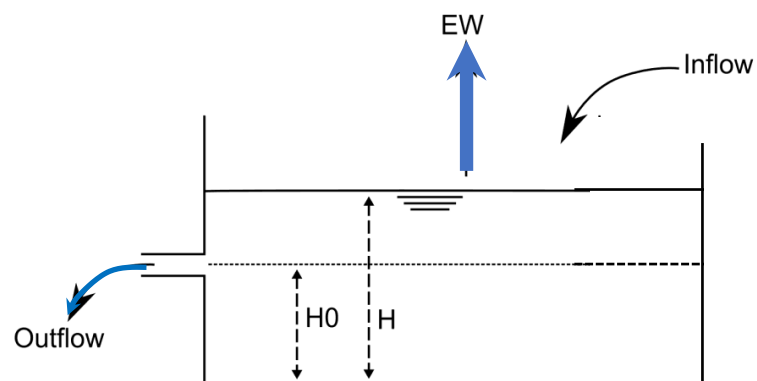


Increasing lake area

- Decreasing water level
- Decreasing outflow
- Increasing evaporation

Decreasing lake area

- Increasing water level
- Increasing outflow
- Decreasing evaporation



Reservoirs –Standard

- A total storage capacity S is assigned to each reservoir, and the fraction of filling of a reservoir is calculated.
- Reservoirs in standard CWatM are managed with different levels of release dependent on relative storage, representing a steady flow. This implementation is continued for reservoirs without user-defined operations.

$$F = \frac{Vt}{S}$$

The levels are defined as:

- a) The “conservative storage limit” fraction because a reservoir should never be completely empty
- b) For prevention of damage in case of flooding, a reservoir should not be filled to the full storage capacity.
- c) the “normal storage limit” defines the buffering capacity and the available storage of a reservoir

Between the state of flood and conservative storage limit, a reservoir is managed as much as possible to deliver a constant outflow so that there is also a constant energy output from hydropower generation

Reservoirs - Standard

Equation of 4 rules:

$$O_{res} = \min(O_{min}, \frac{1}{\Delta t} F \cdot S) \quad F \leq 2L_c$$

$$O_{res} = O_{min} + (O_{norm} - O_{min}) \frac{(F - 2L_c)}{(L_n - 2L_c)} \quad L_n \geq F > 2L_c$$

$$O_{res} = O_{norm} + \frac{(F - L_n)}{(L_f - L_n)} \cdot \max\{(I_{res} - O_{norm}), (O_{nd} - O_{norm})\} \quad L_f \geq F > L_n$$

$$O_{res} = \max(\frac{(F - L_f)}{\Delta t} S, O_{nd}) \quad F > L_f$$

where:

S	: Reservoir storage capacity [m ³]
F	: Reservoir fill (fraction, 1 at total storage capacity) [-]
L _c	: Conservative storage limit [-]
L _n	: Normal storage limit [-]
L _f	: Flood storage limit [-]
O _{min}	: Minimum outflow [m ³ s ⁻¹]
O _{norm}	: Normal outflow [m ³ s ⁻¹]
O _{nd}	: Non-damaging outflow [m ³ s ⁻¹]
I _{res}	: Reservoir inflow [m ³ s ⁻¹]

If storage filling is less equal

2 x conservative filling

-> try to preserve a minimum flow

If storage filling is lower equal normal filling

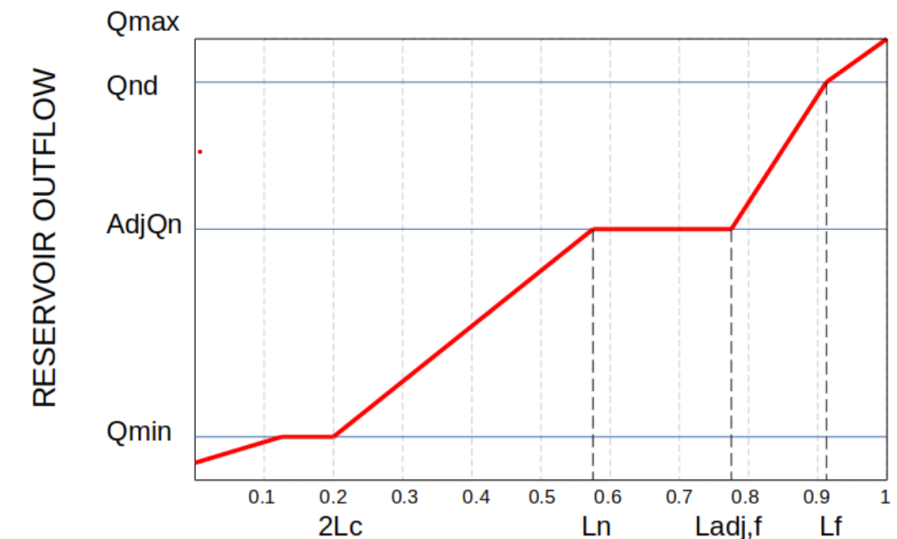
-> try to get close to normal outflow

If storage filling is above normal filling

-> try to get close to normal outflow
(storing water in the reservoir)

If storage filling is above flood storage filling

-> to prevent flood damage



Reservoirs - Standard

Data for reservoirs

- Reservoir volume - lakesResVolRes.nc
- Reservoir type - lakesResType.nc (1 is a lake, 2 is a reservoir)
- Reservoir year - lakesResYear.nc (when is a reservoir activated)

Parameter for reservoirs in the settingsfile

```
# Reservoirs
# reservoir volume from HydroLakes database
waterBodyVolRes = $(PathLakesRes)/lakesResVolRes.nc
# reservoir starting year from HydroLakes database
waterBodyYear = $(PathLakesRes)/lakesResYear.nc
# Conservative, normal and flood storage limit (fraction of total storage, [-])
conservativeStorageLimit = 0.1
#normalStorageLimit = 0.5    # --> put into calibration
floodStorageLimit = 0.9
# adjusting the balance between normal and flood storage
# [0 ..1]  0: NormalstorageLimit      1: (= closer to flood) results in keeping the normal
adjust_Normal_Flood = 0.5

# Minimum, Normal and Non-damaging reservoir outflow  (fraction of average discharge, [-])
MinOutflowQ = 0.2
NormalOutflowQ = 1.0
NonDamagingOutflowQ = 4.0
```

Reservoir module updates - Composing release curves from observations (Smilovic, 2020)

- Reservoirs are associated with “command areas”, defined as the area to which they can serve to satisfy demands.
- Command areas can be input as user-defined maps – otherwise, standard CWatM creates a buffer of user-defined width around all lakes and reservoirs.
- Release curves: Curves are separate for downstream release, satisfying demand, and transfers between reservoirs.
- We present three approaches to compose release curves based on observations.



Reservoir module updates – Approaches

- **Approach 1: Annual pattern in reservoir volumes data observed.**

Release curves calculated from daily observations. Data requirements: daily reservoir storage observation, release fraction and duration.

Reservoir curve calculated from observations.

$$\text{Reservoir Volume}_{\text{day}} = \text{Reservoir Storage}_{\text{day}} \times \text{Release Fraction}_{\text{day}}$$

$$\frac{1}{\text{total release duration} - \text{day of release}}$$

Julian Day	Release fraction
309	0
310	1/146
311	1/145
312	1/144
...	...
86	1/4
87	1/3
88	1/2
89	0

- **Approach 2: Storage and releases**

If time series of reservoir storage and releases are each available, the daily release ratio can be calculated as the average (mean, median, running average, etc.) of the observed release ratios.

$$\text{Ratio}_{\text{day}} = \text{median}_{\text{years}} \left\{ \frac{\text{release day, year}}{\text{live storage day, year}} \right\}$$

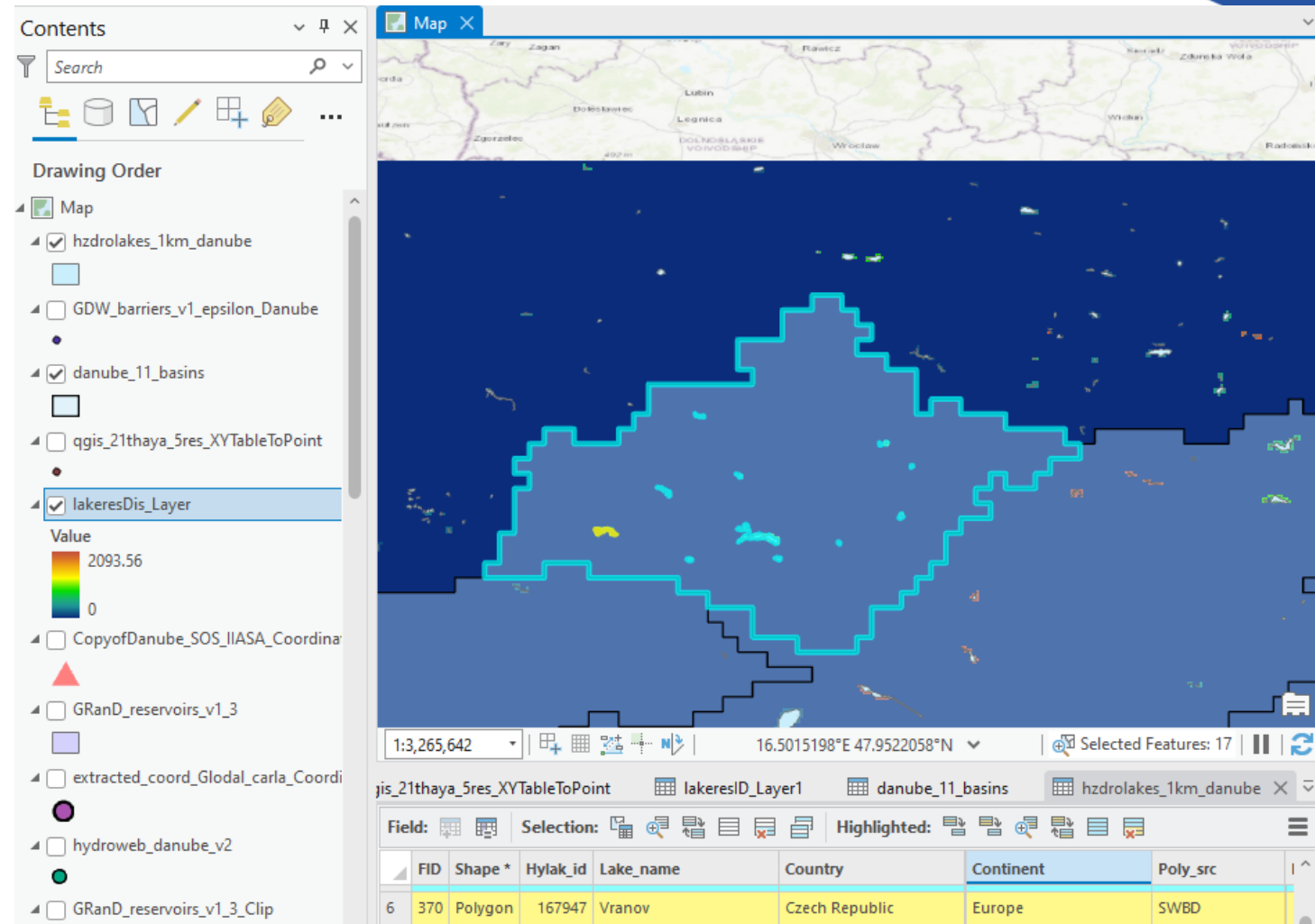
Hands on CWatM

1. Input Data for reservoirs from Hydro Lake database
2. ArcGIS, Panoply or another software able to read netcdf files
3. Your timeseries and input data
4. Excel_settings_reservoir.xlsx

Input from Hydrolakes database

Input Data for reservoirs from Hydrolake database

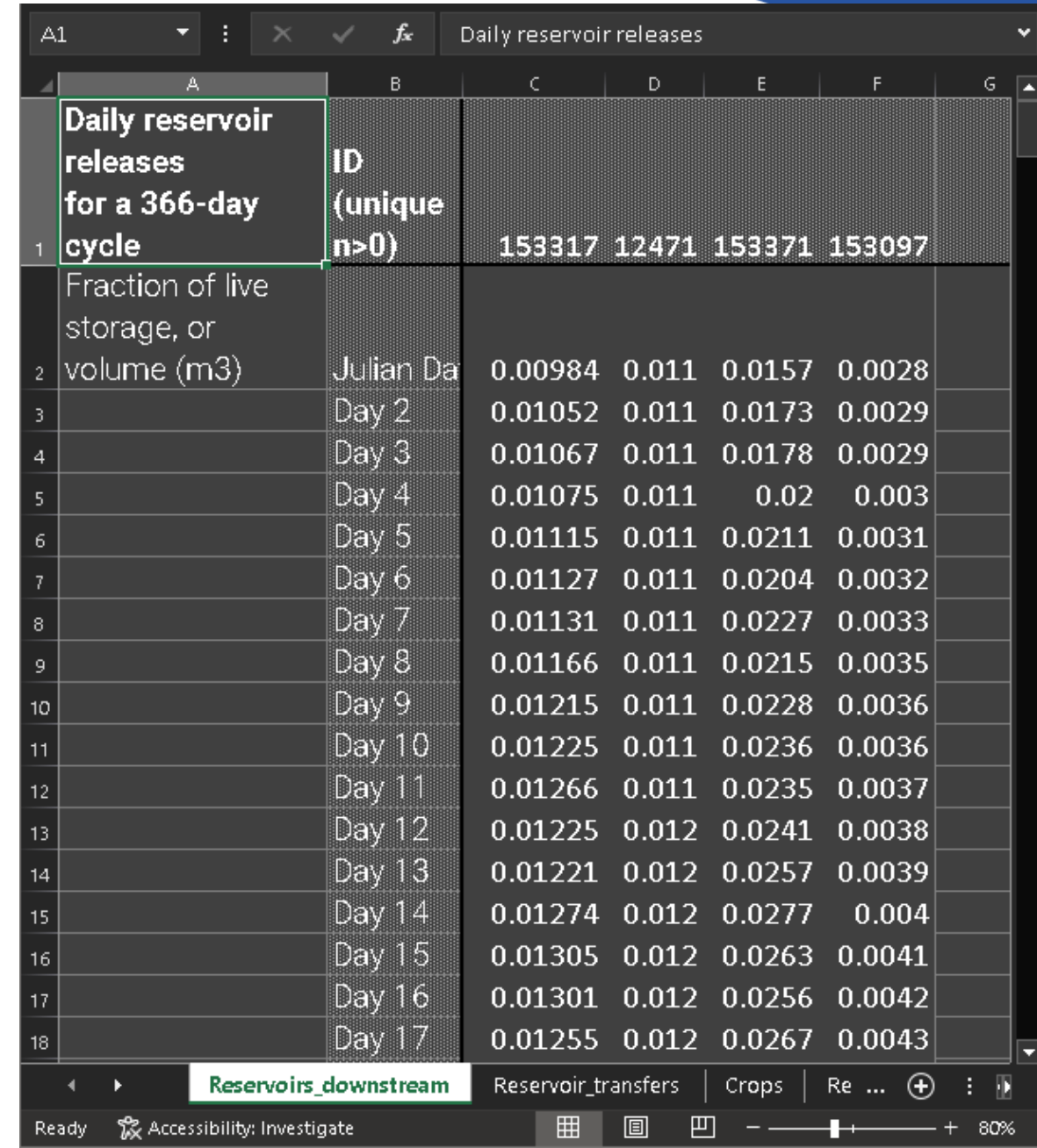
1. Reservoir ID (Hyak ID) – lakeResID.nc
2. Reservoir volume - lakesResVolRes.nc
3. Reservoir type - lakesResType.nc (1 is a lake, 2 is a reservoir)
4. Reservoir year - lakesResYear.nc (when is a reservoir activated)
5. Nominal discharge (optional) – lakeresDis.nc



Reservoir update in CWatM

Reservoirs in models decide at each time step, how much water to release downstream into the river, to satisfy demands, or transfer to another reservoir. We aim to determine multi-objective reservoir releases based on observations providing historically similar releases for future scenarios.

Users input 366-day release curves, either with the fraction of live storage (values ≤ 1) volume in m3 (values > 1) to be potentially release. Sheets *Reservoirs_downstream* and *Reservoirs_supply* share the same format. *Sheet_transfers* is introduced next.

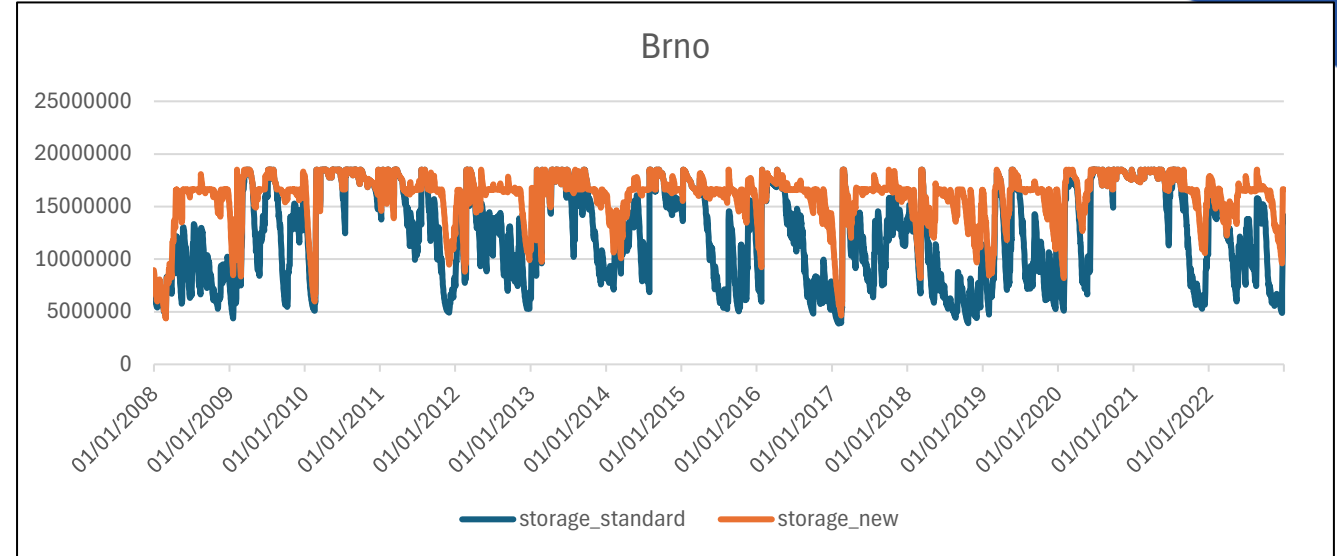


Daily reservoir releases					
Daily reservoir releases for a 366-day cycle	ID (unique n>0)				
		153317	12471	153371	153097
Fraction of live storage, or volume (m3)	Julian Day	0.00984	0.011	0.0157	0.0028
	Day 2	0.01052	0.011	0.0173	0.0029
	Day 3	0.01067	0.011	0.0178	0.0029
	Day 4	0.01075	0.011	0.02	0.003
	Day 5	0.01115	0.011	0.0211	0.0031
	Day 6	0.01127	0.011	0.0204	0.0032
	Day 7	0.01131	0.011	0.0227	0.0033
	Day 8	0.01166	0.011	0.0215	0.0035
	Day 9	0.01215	0.011	0.0228	0.0036
	Day 10	0.01225	0.011	0.0236	0.0036
	Day 11	0.01266	0.011	0.0235	0.0037
	Day 12	0.01225	0.012	0.0241	0.0038
	Day 13	0.01221	0.012	0.0257	0.0039
	Day 14	0.01274	0.012	0.0277	0.004
	Day 15	0.01305	0.012	0.0263	0.0041
	Day 16	0.01301	0.012	0.0256	0.0042
	Day 17	0.01255	0.012	0.0267	0.0043

	A	B	C	D	E	F	G	H	I
1									
2			GUIDE		Giving reservoir ID	153097			
3			Reservoir transfers, inwards or outwards, are activated when reservoir_transfers = True in the text settings file. Choose the waterbody ID of the giving and receiving reservoirs and the amount, either as a fraction of live storage (values ≤ 1) or in cubic metres (values > 1). Waterbody ID 0 represents the infinite outside source/sink. Outward transfers are added to industrial demand/use.		Receiving reservoir ID	153371			
4			Add or remove columns as necessary.		Julian day	Fraction of live storage (values ≤ 1) or metres ³ (values > 1)			
5			Reservoir transfers occur daily, with Giving reservoirs transferring water to associated Receiving		Day 1	0.9			
6			reservoirs. The amount of water transferred is determined by the associated fraction of live storage of the		Day 2	0.9			
7			Giving reservoir, and the available space in the Receiving reservoir. If the Giving reservoir ID is 0		Day 3	0.9			
8			(Representing an outside source/sink), the water amount refers to the Receiving reservoir. Reservoirs can		Day 4	0.9			
9			both receive and give to several reservoirs, and transfers are executed in the order they appear in the		Day 5	0.9			
10			Excel sheet.		Day 6	0.9			
11					Day 7	0.9			
12					Day 8	0.9			
13					Day 9	0.9			
14		OUTPUT VARIABLES	UNITS = CUBIC METRES		Day 10	0.9			
15		reservoir_transfers_net_M3	net storage change of reservoirs after daily transfers		Day 11	0.9			
16		reservoir_transfers_in_M3	water received into reservoirs		Day 12	0.9			
17		reservoir_transfers_out_M3	water given from reservoirs		Day 13	0.9			
18		reservoir_transfers_from_outside_M3	water received into reservoirs from Outside		Day 14	0.9			
19		reservoir_transfers_to_outside_M3	water given from reservoirs to the Outside		Day 15	0.9			
20					Day 16	0.9			

Some results

- Orange line shows storage changes with the implementation new version.
- Blue line is the standard CWatM storage.



Reservoir	Correlation (new)
Vranov	0.63
Balaton	0.51
Bistrita-7	0.68
Vir	0.64

